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Digital Electronics & VLSI
Codec Technologies India
Internship 2026

Digital Voting Machine with Secure Memory

Debounce ballot FSM, CRC-8 records in an I2C EEPROM with write-verify, tamper seal, PIN-gated 7-segment results, HD44780 LCD

AT A GLANCE

32 / 32 checks PASS

System test
4 written, 4 verified,
CRC ok

EEPROM records

8 glitches -> 1 vote

Bounce rejection
1467 LC (19 %), 88.29
MHz

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1020 LUT4, 704 FF

Yosys synth_ice40

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Xilinx xc7 (stat)

matches datasheet

LCD init sequence

seq16 | cand | CRC-8

Record format

VERDICT

SIM PASS CRC VERIFIED

SYNTH OK TIMING MET

NO HW RUN

VIVADO NOT RUN

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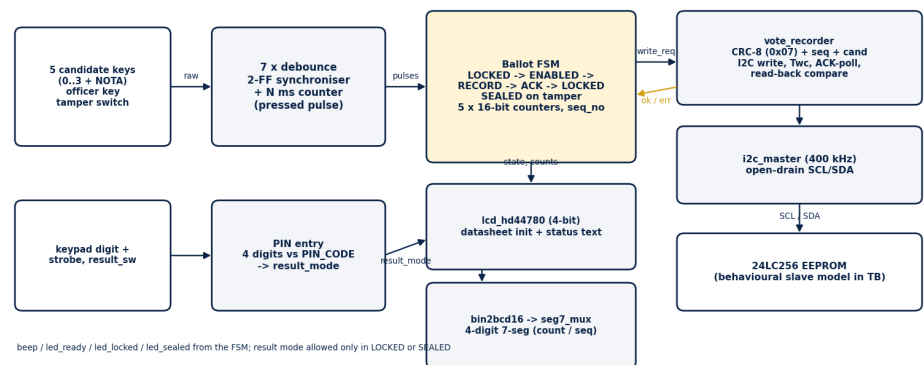
Objective and brief mapping

The brief asked for a Verilog voting machine: 4 candidates + NOTA, parameterised debouncers with synchronisers, an officer-enabled single-vote ballot FSM with beep/LED acknowledge, 16-bit counters, secure storage of each vote in a 24LC256 I2C EEPROM (own slave model) with CRC-8, monotonic sequence number and write-verify, a tamper seal, PIN-gated results on a 4-digit multiplexed 7-segment display, a HD44780 4-bit LCD driver whose init sequence is checked against the datasheet, a testbench with bounce injection, waveform PNGs, and iCE40 + Xilinx synthesis with Vivado scripts declared not run. All of it was built and simulated here; nothing ran on hardware.

Architecture

Ballot FSM state	Entered on	Behaviour
LOCKED	reset, end of ACK	candidate keys ignored; officer key -> ENABLED (not in result mode)
ENABLED	officer pulse	first debounced candidate pulse: counter++, seq_no++, record request -> RECORD
RECORD	vote accepted	further presses ignored; wait for recorder verify_ok / verify_err
ACK / ERROR	verify result	beep 500 ms then LOCKED / chirp until officer re-enables
SEALED	tamper level, any state	no votes ever again; results still readable with PIN; display shows ----

voting_machine -- 4 candidates + NOTA, secure I2C EEPROM record store, PIN-gated results



Implementation details

- debounce.v: 2-flop synchroniser, counter restarts on agreement, flips after DEBOUNCE_MS (20 ms at 12 MHz = 240 000 cycles), one-clock pressed pulse.
- vote_recorder.v: CRC-8 (poly 0x07) over seq_hi, seq_lo, cand; page write START A0 addr_hi addr_lo d0..d3 STOP; wait Twc; ACK-poll the device address; word address; repeated START in read mode; 4 bytes (last NACKed); byte compare -> verify_ok/err. Record address = seq * 4, so 32 KB holds 8191 ballots.
- i2c_master.v: quarter-bit state machine, open-drain scl_oe/sda_oe, START/repeated START/STOP generation, rx_ack capture (the NACK path is what ACK polling uses).
- PIN: 4-digit shift register compared with PIN_CODE on the 4th digit; result_mode = result_sw & pin_ok & (LOCKED | SEALED). Candidate keys select the displayed count; bin2bcd16 (16-cycle double dabble) feeds the 4-digit multiplexed 7-segment.
- lcd_hd44780.v: 15 ms, nibbles 3 3 3 2, bytes 28 08 01 06 0C, then 80 + 16 chars from a 6-message ROM; rewritten whenever the state message changes.
- tb/i2c_eeprom_24lc256.v: my behavioural slave (32 KB, 64-byte page buffer committed at STOP, Twc busy -> NACK, sequential read with auto-increment, START/STOP/byte/NACK counters).

Verification (Icarus Verilog 14; sim-scaled: 1 MHz clock, 2 ms debounce, 100 kHz I2C, Twc 300 us vs model 500 us)

Checks	What was tested	Result
1-5	LCD init nibbles 3,3,3,2; bytes 28 08 01 06 0C; RS=0 commands, 0x80 then RS=1 data; first line 'BALLOT LOCKED'	PASS
6-10	press without enable ignored; officer key -> led_ready; bouncy candidate-1 press (8 glitches) -> count 1, record 1 verified, 1 beep, lock	PASS
11-13	candidate 2 then 3 in one ballot -> only 2; press while locked rejected; NOTA and 2nd cand-1 vote -> counts 1/2, seq 4, 4 records ok	PASS
14-23	EEPROM via slave model: 00 01 01 12 / 00 02 02 24 / 00 03 04 23 / 00 04 01 53, seq monotonic, CRC-8 valid x4, 16 bytes in 4 page writes, NACKs seen during Twc (20 STARTs / 8 STOPs)	PASS
24-29	PIN 1235 rejected, 1234 accepted; 7-seg shows 0002 for candidate 1 with digit pattern '2'; officer key inert in result mode; mode ends with switch	PASS
30-32	tamper -> SEALED, no votes after seal, results still readable with PIN	PASS

KEY TAKEAWAY Every vote becomes a CRC-protected, sequence-numbered EEPROM record that is read back before the machine beeps; bounce, double presses, tamper and PIN gating are all covered by the 32-check testbench.

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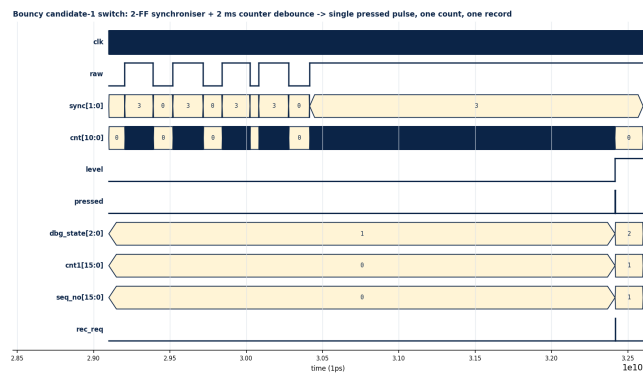
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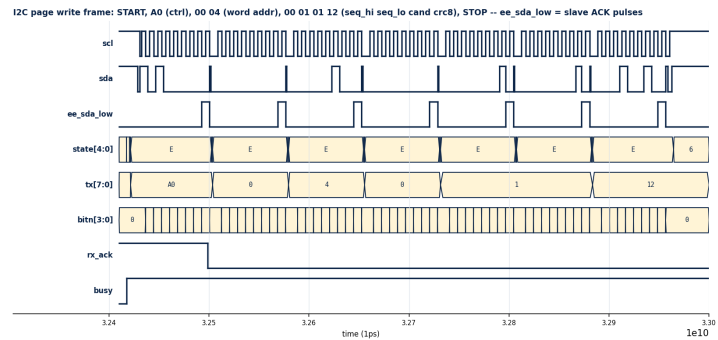
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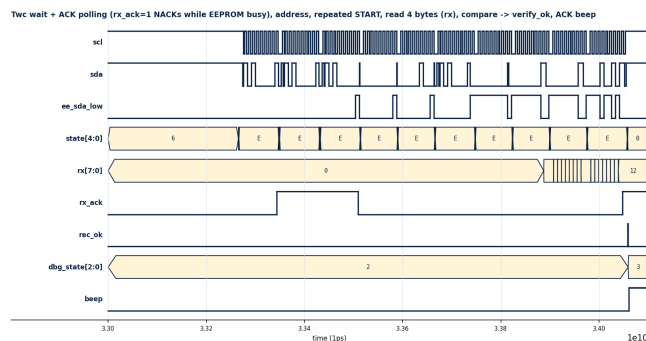
Bouncy candidate-1 key: raw glitches, 2-flop sync, debounce counter, one pressed pulse, cnt1 and seq_no increment once, rec_req fires once.

I2C record write and read-back verify



Page write frame: START, A0, 00 04, 00 01 01 12, STOP; ee_sda_low pulses are the slave's ACKs (rx_ack drops to 0 after the control byte).

Verify phase and result display



Twc wait, ACK polling (rx_ack = 1 while the EEPROM is busy), word address, repeated START, 4 bytes read into rx (last = CRC 0x12), compare -> rec_ok, FSM to ACK, beep.

Synthesis and implementation results

Flip-flops dominate: seven debounce counters plus the LCD, Twc and acknowledge timers. Narrowing those timers from 32 to 24 bits removed 97 iCE40 cells (2145 -> 2048 incl. scope markers). The 143 I/O are mostly observability ports for the testbench (counters, seq_no, state).

Flow (run here)	Measured result
Yosys 0.68 synth_ice40	2035 cells: 1020 SB_LUT4, 704 FF (395 DFFESR, 193 DFFSR, 56 DFFE, 35 DFFESS, 13 DFF, 12 DFFSS), 311 SB_CARRY
nextpnr-ice40 hx8k ct256, --freq 12 --seed 1	1467 / 7680 LC (19 %), 143 SB_IO; Fmax 88.29 MHz, PASS at 12 MHz
icetime -d hx8k	total path delay 11.89 ns (84.11 MHz)
Yosys synth_xilinx xc7 (stat)	1593 cells: 612 LUT2-6, 723 FF (676 FDRE + 47 FDSE), 83 CARRY4, 31 INV, 16 IBUF, 127 OBUF
Per-module FF / LUT (synth_xilinx hierarchy)	7 x debounce 22 FF / 10 LUT each; vote_recorder 125 / 133; i2c_master 43 / 40; lcd_hd44780 62 / 121; bin2bcd16 63 / 38; seg7_mux 25 / 15; crc8 0 / 10; top (FSM, counters, PIN, display mux) 251 / 185 -> total 723 FF / 612 LUT

EEPROM image after the test (read from the slave model) and PIN-gated result display

The sequence number doubles as ballot count and record address (seq x 4): an auditor sees a missing or duplicated ballot as a sequence gap and a modified byte as a CRC-8 mismatch, and the read-back means the machine only beeps once the record is physically in the EEPROM. Slave-model counters: 4 page writes of 4 bytes, 8 STOPs, 20 STARTs (ACK-polling retries plus repeated STARTs), 0 verify

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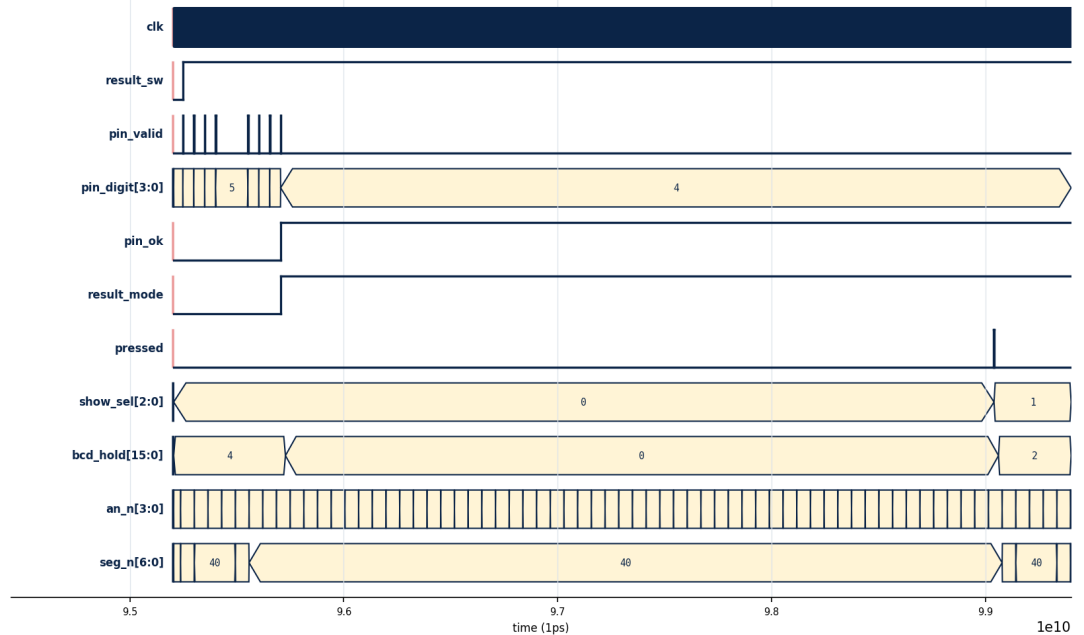
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errors. 32 KB holds 8191 records.

Ballot (seq)	EEPROM address	Bytes stored	CRC-8 recomputed in TB	Candidate
1	0x0004	00 01 01 12	0x12 = match	1
2	0x0008	00 02 02 24	0x24 = match	2
3	0x000C	00 03 04 23	0x23 = match	4 (NOTA)
4	0x0010	00 04 01 53	0x53 = match	1

Result mode: wrong PIN 1235 rejected, PIN 1234 accepted, candidate-1 button selects count 0002 on the 7-segment mux



Result mode: four digits 1,2,3,5 leave pin_ok low; 1,2,3,4 raise pin_ok and result_mode; the candidate-1 key selects show_sel = 1 and bcd_hold becomes 0002, multiplexed onto seg_n / an_n.

HD44780 command sequence captured on EN falling edges (sim/voting_sim.log) vs datasheet 4-bit initialisation

Captured (RS:nibble)	Byte	Datasheet step	Wait after
0:3 0:3 0:3	0x30 x3	Function set 8-bit, sent three times after power-on (>15 ms, >4.1 ms, >100 us)	4.1 ms / 100 us / 100 us
0:2	0x20	Function set: switch to 4-bit interface	100 us
0:2 0:8	0x28	Function set: 4-bit, 2 lines, 5x8 font	100 us
0:0 0:8	0x08	Display off	100 us
0:0 0:1	0x01	Clear display	1.64 ms
0:0 0:6	0x06	Entry mode: increment, no shift	100 us
0:0 0:c	0x0C	Display on, cursor off, blink off	100 us
0:8 0:0	0x80	Set DDRAM address 0 (line 1)	100 us
1:4 1:2 ... (16 bytes)	'B' 'A' ...	Data writes: BALLOT LOCKED + 3 spaces (565 nibbles captured over the whole test = init + 15 status rewrites)	-

Declared substitutions / not run here

- No hardware: no switches, 24LC256, HD44780, buzzer or FPGA board. The EEPROM is my behavioural slave model written from the datasheet protocol; the LCD is verified at command-sequence level only.
- Vivado not available: synth/vivado_vm.tcl and synth/vm_basys3.xdc (Basys-3, 100 MHz) are provided and were not run; Xilinx figures are Yosys synth_xilinx estimates.
- nextpnr constrained 32 real pins (synth/vm.pcf) and auto-assigned the segment lines and debug ports; synth/vm.asc was not programmed. Open-drain I2C is modelled as scl_oe / sda_oe / sda_in and needs a tri-state wrapper on a board. Simulation used Icarus Verilog, not ModelSim.

Learnings / next steps

Two things bit me: a loop variable shared by a combinational and a clocked block produced a constant-driver conflict in Yosys (fixed with separate variables), and the ACK-polling path was invisible until the EEPROM model was made slower than the recorder's Twc wait. CRC-8 detects corruption, not forgery: the next version should MAC each record with a key held inside the FPGA, add a real-time-clock timestamp, and dump the results over UART for the counting centre. A second officer PIN for the counting agent and a display of the total ballot count during voting would complete the EVM feature set.

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